

MITCHELL GRAY

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Education

Georgia Tech, Atlanta, GA

Master's of Science, Computer Science

Jan 2025 – Aug 2026 (Part-Time)

GPA: 4.0; Artificial Intelligence

Cornell University, Ithaca, NY

Bachelor's of Science, Computer Science

Aug 2020 – May 2024

Distributed Systems

Minor, Operations Research Information Engineering

30 Extra Graduate-Level Credits

Knowledge Domain: Distributed Systems · Systems Programming · AI · Machine Learning · Databases · Cloud

Technical Skills

Languages: C · C++ · Python · PL/SQL · Java · Rust

Software & Tools: Git · Pytorch · Scikit-learn · NumPy · Pandas · Poetry · CMake · CI/CD · Docker · K8s · Azure · AWS

Relevant Work Experience

Oracle

Redwood City, CA

Senior Software Engineer | C, PL/SQL, Raft, Java, OracleDB, OCI

Sept 2026 – Present

- Working on Raft Replication in the Oracle Globally Distributed Database for Shard Replication · Writing fault-tolerant and scalable Distributed Systems code in **C**, **Java**, and **PL/SQL**.
- Designed and implemented a generic shard-operations replication framework that consolidated multiple ad-hoc code paths into a single framework · Improved maintainability, scalability, and ease of creating new operations with fewer defects.

Software Engineer II

Sept 2025 – Sept 2026

- Designed and implemented a generic shard-operations replication framework that consolidated multiple ad-hoc code paths into a single framework · Improved maintainability, scalability, and ease of creating new operations with fewer defects.
- Provided technical mentorship and guided the team's newest engineer · Accelerated onboarding and knowledge transfer.
- Designed and implemented a generic shard-operations replication framework that consolidated multiple ad-hoc code paths into a single framework · Improved maintainability, scalability, and ease of creating new operations with fewer defects.
- Provided technical mentorship and guided the team's newest engineer · Accelerated onboarding and knowledge transfer.

Software Engineer I

June 2024 – Sept 2025

- Engineered a fault-tolerant replica replication feature on Oracle Cloud Infrastructure · Enforced placement of Raft replicas on separate server racks to eliminate single-rack replication-unit failures and improve database availability.
- Refactored over 1,200 lines of Sharding serialization code to eliminate a critical data-loss defect · Designed robust de/serialization pipelines and multi-format conversion logic that greatly improved data integrity and system reliability.
- Optimized multiple replica-to-replica data transfer operations · Achieved a **700%** performance improvement without workload and **250%** improvement under active workloads after optimization code changes.
- Resolved critical recoverability issue in replica-to-replica data transfer operations by ensuring complete transfer of essential recovery-related tables and data between replicas · Improved reliability and system integrity during failure.
- Implemented comprehensive Pluggable DBs lifecycle management · Enabled seamless Drop, Unplug, and Plug operations of PDBs to streamline maintenance for Raft replicas using Oracle's multi-tenant container architecture.
- Enhanced error-handling mechanisms throughout the codebase · Improved error handling reliability and diagnosability.

Gecko Robotics

Pittsburgh, PA

Software Engineer Intern | C++, Python, C, CMake, GCP, CI/CD

May 2023 – Aug 2023

- Worked on the Robot Controls team · Wrote code for an asynchronous distributed system in **C++**, **Python**, and **C**.
- Revamped Robot & DAQ emulators · Implemented communications protocol and Client/Server TCP networking.
- Replaced VS build-system with **CMake** · Added calibration support to emulators · Integrated **CI/CD** and **Poetry**.

Carnegie Mellon University - Software Engineering Institute

Pittsburgh, PA

DevOps Engineer Intern | Python, Rust, CI/CD, Docker, Kubernetes

May 2022 – May 2023

- Developed **Python**, **Rust**, and **Bash** code on various NDA projects · Updated and created **CI/CD** pipelines · Created GitLab **REST API** data visualizations for clients · Used ArgoCD and Helm to deploy AWS EKS cluster.
- Improved efficiency of the company by using **Python**, **Neo4j**, NeoDash, Confluence Documents, and the PageRank algorithm to create scripts, metrics, and long-term documentation that are now used throughout the company.

Relevant Academic Work Experience

Georgia Tech Contextual Computing Group

Atlanta, GA

Graduate Teaching Assistant | **AI, ML**

June 2025 – Aug 2026

- Worked on Nosi, an IDE that utilizes **Machine Learning** to detect cheating students · Published in **Learning at Scale 2026** and **AI Ed 2026** · Integrated the IDE in multiple Georgia Tech Graduate CS courses · Sold our solution to multiple companies including a **Fortune 50** company.

Georgia Tech

Atlanta, GA

Graduate Teaching Assistant | **AI, Python, NumPy**

May 2025 – Aug 2026

- Taught Graduate **Artificial Intelligence** · Held Office Hours weekly · Created exams questions.

Cornell University

Ithaca, NY

Teaching Assistant | **Python, NumPy, SQL, Java**

Aug 2022 – May 2024

- Received the **Course Staff Exceptional Service Award** · Taught Databases, Object-Oriented Programming, and Physics · Lectured 400+ students with custom-designed lecture content · Led Exam sessions and other TAs.

Cornell Database Group

Ithaca, NY

Database Research Intern | **Java, AWS**

Jan 2022 – May 2024

- Published and presented paper at **VLDB 2023** combining **Reinforcement Learning** and Worst-Case Optimal Joins.
- Worked on upgrading query engine to a **Distributed Architecture** · Created dynamic statistics & RL visualizations.

Publications

AI ED [link]

2026

Publication

Learning at Scale [link]

2026

Publication

VLDB [link]

2023

Publication

Awards

Oracle

Date 2026

Mentorship and Leadership Award

Cornell Bowers CIS

May 2024

Course Staff Exceptional Service Award

Projects

Distributed Testing Platform (SPEED) | Masters Project | **Java, JUnit, Cloud** | **Github** Aug 2023 – May 2024

- My team and I created a **Scalable Platform for Efficient Execution of Distributed testing**. The fault-tolerant system contains a controller node that orchestrates worker nodes that run **JUnit** tests on **Java** code. The worker nodes report their findings to the controller. Once all tests are ran, test results are shown in the frontend to the user.

DS-Labs Sharded Key Value Store | Distributed Computing | **Java, Paxos, 2PC** | **Github** Jan 2023 – May 2023

- Using the DS-Lab framework, my partner and I were able to create a sharded transactional key-value store that uses **Paxos** for replication and **2PC** for multi-key updates. We implemented an Exactly-once RPC protocol on top of an asynchronous network, a primary-backup protocol, Paxos implemented with the PMMC protocol, and 2PC.

Cornell Meetup | Cloud Computing | **Azure, CosmosDB, Python, WebDev** | **Github** | **BOOM** Aug 2022 – Dec 2022

- 1 of 32 projects selected for BOOM 2023. My partner and I created a social media web app that allows users to create groups, chat with friends, and see where their friends are when on campus. Accounts details were obfuscated and salted.